

National Institute of Technology, Kurukshetra
Department of Robotics and Industrial Engineering
Theory Examination (End-Term) - Question Paper

Month and Year of the Examination:

Aug 2026 (13-08-2026)

Programme: B. Tech. (Prodⁿ & Industrial Engg.)

Semester: Fourth (4th) Sem

Subject: Materials Science

Max. Marks: 50

Course Code: PIPC-215

Time allowed: 03 hours

Note: Attempt ALL FIVE question. State clearly the assumptions made wherever required. Assume any suitable data if not given and highlight it. Steps and procedure are important, do not omit them, these are considered in rubrics and carries marks.

Que.1-a): Calculate the linear density and planer density for FCC unit cell. (CLO-1, 5 marks)

Que.1-b): Elaborate with your understanding the point, line and surface defect. (CLO-1, 5 marks)

Que.2): Draw and explain the Fe-C diagram with all phases and micro-constituents, invariant reactions, and critical temperatures. Also, calculate the phase fractions for 0.65 wt % C with its microstructure development at pre and post-austenite stage. (CLO-1, 10 marks)

Que.3-a): Elaborate with neat sketch and practical significance for the strain hardening with respect to stress-strain curve. (CLO-2, 5 marks)

Que.3-b): Derive the Lever Rule, also draw the neat sketch and explain its significance. (CLO-2, 5 marks)

Que.4-a): Explain the 'Glow Discharge Nitriding' using neat sketch. State the limitations of the process. (CLO-3, 5 marks)

Que.4-b): Describe the method of 'Flame Hardening' in details along with its sub-types. (CLO-3, 5 marks)

Que.5-a): Write a short note on 'shape memory alloy'. (CLO-4, 5 marks)

Que.5-b): What do you understand by 'Annealing of Steel'? Explain its all types using neat sketches. (CLO-4, 5 marks)

----- End of Question paper -----

*****All The Best*****